

Scaling Up Agricultural Insurance

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Agriculture is Risky

Long and rich tradition in development economics studying effects of risk, especially in agricultural settings (Paxson 1992; Rosenzweig and Binswanger 1993; Udry 1994, 1995; Mobarak and Rosenzweig 2013; Donovan 2021)

- Risk lowers utility because households prefer stable consumption
- Risk can lower output if high-return investments are risky
- This may explain apparent underutilization of inputs and inefficient crop choice

A potential solution is to offer insurance

- Microeconomic research studying effects of insurance (Karlan et al. 2014)

In parallel, macroeconomic work thinking about GE effects of scaling up interventions

- e.g. Buera et al. (2021), Lagakos et al. (2023), Bergquist et al. (2025),
+ many papers at this conference!

Insurance Unlocks Investment (Karlan et al. 2014)

Our paper begins with the RCT of Karlan, Osei, Osei-Akoto, and Udry (2014)

- Study rainfall insurance, which pays out if there is not enough rain (or too much)
 - Sidesteps moral hazard/adverse selection: rain is exogenous and common knowledge
- Randomize grants of insurance, and insurance prices
- Insurance raises farmer's agricultural investment
- Insurance shifts crop choice towards risky, high-return crop (maize vs. mangoes)

Strong evidence that risk aversion plays a role in constraining farmers

Our Paper: Scaling Up Agricultural Insurance

Should government “scale up” agricultural insurance?

- In practice, this means subsidizing it or offering it at below-market cost

Experiment identifies a partial equilibrium effect:

- Compares treated and control households within the same economy
- If scaled up, there will be important GE effects

To determine optimal policy, we need to understand:

- Is there a market failure?
- *If so, does subsidizing insurance fix the incompleteness?*

Outline of the Talk

To answer these questions:

- **Model:** Agricultural households that can (imperfectly) share risk
- **Optimal Policy:** Sufficient-statistics decomposition of how subsidy affects welfare
- **Calibration:** Use experimental moments, plus additional data
- **Counterfactuals:** What happens when we subsidize insurance?

We find that:

- The optimal subsidy trades off three terms:
 - **Fiscal Externality:** Classic distortion from subsidizing insurance below cost
 - **Pecuniary Externality:** Change in crop prices transfers resources across states
 - **Fiscal Incidence:** Redistribution from insurance buyers to non-buyers
- **Quantitatively, optimal subsidy is small and negative:**
 - Optimal subsidy is -0.09% of cost of insurance

Model

Environment and Utility

Continuum of ex-ante identical regions, $i \in [0, 1]$

Representative household in each region, all agents price takers, competitive markets

In each region, s_i^H is good state (rain) and s_i^L is bad state (drought)

States are drawn iid across regions, so no aggregate uncertainty

Household utility in region i :

$$\begin{aligned} U_i &= \mathbb{E} [u(c_i(s_i))] \\ &= P(s_i^H) \cdot u(c_i(s_i^H)) + P(s_i^L) \cdot u(c_i(s_i^L)) \end{aligned}$$

For quantitative model, $u(c) = \frac{c^{1-\gamma}-1}{1-\gamma}$ is CRRA

Farm Production

Household makes income by farming; G different crops, indexed by g

Before knowing state, allocates land across crops and chooses fertilizer:

$$y_{g,i} = y_{g,i}(s_i, l_{g,i}, f_i)$$

$$\sum_{g=1}^G l_{g,i} = 1$$

$$f_i \geq 0 \text{ and } l_{g,i} \geq 0$$

For quantitative model, $y_{g,i}(s_i, l_{g,i}, f_i) = z_g(s_i) \cdot l_{g,i}^{\theta_g} f_i^\phi$

Curvature θ_g reflects that plots have heterogeneous suitability

- As more land allocated to maize, marginal land less suited to maize

Agricultural Profits

Farmer takes prices as given, agricultural profits are:

$$\pi_i \left(s_i, \{l_{g,i}\}_{g=1}^G, f_i \right) = \sum_{g=1}^G p_g \cdot y_{g,i} (s_i, l_{g,i}, f_i) - f_i$$

Crop choice entails risk-return tradeoff

- For farmers in Karlan et al. (2014), maize has higher expected return, but mangoes are drought-resistant

Agricultural Insurance

Each household can buy insurance, x_i

- Costs $p_{x,i} \cdot x_i$, paid regardless of state
- Pays out x_i in the bad state, 0 in good state
- For simplicity, constrain $x_i \geq 0$ (household doesn't sell insurance)

Household's Problem

Household also receives dividend, Π^I , from an insurance firm

Household budget constraint is:

$$c_i(s_i) = \pi_i(s_i^H) + \Pi^I - p_{x,i} \cdot x_i + 1(s_i = s_i^L) \cdot x_i$$

Each household takes prices and dividends as given and solves:

$$\max_{\{l_{g,i}\}_{g=1}^G, f_i, x_i} \mathbb{E} \left[u \left(\pi_i(s_i) + \Pi^I - p_{x,i} \cdot x_i + 1(s_i = s_i^L) \cdot x_i \right) \right]$$

subject to

- $\pi_i(s_i) = \sum_{g=1}^G p_g \cdot y_{g,i}(s_i, l_{g,i}, f_i) - f_i$
- $\sum_{g=1}^G l_{g,i} = 1$
- $x_i \geq 0, f_i \geq 0$, and $l_{g,i} \geq 0$

Extensive Margin

In practice, many households buy zero insurance

To rationalize this, extend model to have a participation margin:

- Draw utility costs ε_i^0 and ε_i^1
- V^1 is indirect utility when chooses insurance
- V^0 is indirect utility when constrained to zero insurance
- Household gets $V^0 + \bar{\varepsilon} + \varepsilon_i^0$ if it does not participate
- Household gets $V^1 + \varepsilon_i^1$ if it does participate

In quantitative model, $(\varepsilon_i^0, \varepsilon_i^1)$ drawn iid from Type I EV with scale χ

$$P(V^1 + \varepsilon_i^1 > V^0 + \bar{\varepsilon} + \varepsilon_i^0) = \frac{\exp(\chi V^1)}{\exp(\chi V^1) + \exp(\chi(V^0 + \bar{\varepsilon}))}$$

We will show results both with and without extensive margin

Final Good and Fertilizer Sector

Agricultural goods are aggregated, constant-returns-to-scale, into final good

Use CES for quantitative model:

$$Y_g = \int_0^1 y_{g,i}(s_i) di \forall g$$
$$Y = \left(\sum_{g=1}^G Y_g^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}$$

Final good is numeraire ($P = 1$). Competitive pricing:

$$p_g = \frac{dY}{dY_g} = \left(\frac{Y_g}{Y} \right)^{-1/\sigma}$$

Fertilizer sector is also competitive, with linear technology

- Equivalently, fertilizer imported + small open economy
- Normalization: final good converted one-to-one to fertilizer

Insurance Firm

Insurance is sold by a representative insurance firm

- Actuarially fair price of insurance is $P(s_i^L)$
- Firm pays per-unit service cost τ , and earns markup μ

Firm thus offers prices:

$$p_{x,i} = P(s_i^L) + \tau + \mu \forall i$$

and makes profits

$$\Pi^I = \int_0^1 \left(p_{x,i} - P(s_i^L) - \tau \right) \cdot x_i di = \mu \cdot \int_0^1 x_i di$$

Firm is owned by households, so profits are rebated to households

Note that τ and μ are both frictions, but they are distinct:

- Service cost τ is a resource cost; can't just undo it
- Markup μ is a pure wedge; can undo it with a subsidy

Separation and Efficiency

Household first order conditions:

$$\text{Production: } \mathbb{E} \left[\frac{u'(c_i(s_i))}{\mathbb{E}[u'(c_i)]} \cdot p_g \frac{\partial y_{g,i}}{\partial l_{g,i}} \right] = \lambda_i \forall g$$

$$\text{Insurance: } \mathbb{E}[u'(c_i)] \cdot p_{x,i} = u'(c_i(s_i^L)) P(s_i^L)$$

As long as $x_i > 0$, we get a nice separation result:

$$\mathbb{E} \left[\frac{u'(c_i(s_i))}{\mathbb{E}[u'(c_i)]} \cdot p_g \frac{\partial y_{g,i}}{\partial l_{g,i}} \right] = p_{x,i} \cdot p_g \frac{\partial y_{g,i}(s_i^L)}{\partial l_{g,i}} + (1 - p_{x,i}) \cdot p_g \frac{\partial y_{g,i}(s_i^H)}{\partial l_{g,i}}$$

- Farm maximizes marginal-utility-weighted profits
- With insurance, weight by state-prices, $p_{x,i}$ and $1 - p_{x,i}$
- With actuarially fair insurance, maximize (true) expected profits

Distortion in insurance market leads to production inefficiency

Welfare and Optimal Policy

Subsidies as Markups

To scale up agricultural insurance, government provides a subsidy

Subsidy lowers the price consumer pays, and is funded through lump-sum taxation

Markups work in exactly the same way!

- Markup raises price, and profits returned lump-sum to households as dividend

For simplicity, we will lump them together:

- Using subsidies as a tool, government chooses the markup

$$\mu = \mu^{\text{Baseline}} - \text{Insurance Subsidy}$$

In our application, we will assume $\mu^{\text{Baseline}} = 0$, so can think of $\mu = -\text{Subsidy}$

Indirect Utility and Consumption

How do subsidies affect household welfare, in general equilibrium?

- Differentiate welfare with respect to the markup
- Household treats prices and insurer profits as exogenous
- Land and insurance are choices, so fall out with Envelope Theorem

$$\begin{aligned}\frac{dU}{d\mu} &= \frac{\partial U}{\partial p_x} \cdot \frac{dp_x}{d\mu} + \frac{\partial U}{\partial \Pi^I} \cdot \frac{d\Pi^I}{d\mu} + \sum_{g=1}^G \frac{\partial U}{\partial p_g} \cdot \frac{dp_g}{d\mu} \\ &= \mathbb{E} \left[u'(c) \cdot \left(\underbrace{\frac{\partial c}{\partial p_x} \cdot \frac{dp_x}{d\mu}}_{\text{Cost of Insurance}} + \underbrace{\frac{\partial c}{\partial \Pi^I} \cdot \frac{d\Pi^I}{d\mu}}_{\text{Insurer Profits}} + \underbrace{\sum_{g=1}^G \frac{\partial c}{\partial p_g} \cdot \frac{dp_g}{d\mu}}_{\text{Farmer Profits}} \right) \right]\end{aligned}$$

Note: $\frac{d}{d\mu}$ reflects the derivative of a variable in general equilibrium

Optimal Subsidies (Model Without Extensive Margin)

At the optimal subsidy (for model without extensive margin):

$$\underbrace{-\mu \cdot \frac{dx}{d\mu}}_{\text{Fiscal Externality}} = \underbrace{\mathbb{E} \left[\frac{u'(c)}{\mathbb{E}[u'(c)]} \cdot \left(\sum_{g=1}^G y_g \cdot \frac{dp_g}{d\mu} \right) \right]}_{\text{Pecuniary Externality}}$$

- When $\mu = 0$, marginal benefit to consumers from lower p_x exactly cancels marginal cost to the insurer/government
- When $\mu < 0$ (positive subsidy), subsidy imposes a fiscal externality
- Pecuniary externality: change in crop prices transfers resources across states
- Formula very similar to Baily-Chetty: balance fiscal externality against risk term

Pecuniary Externality Reduces Consumption Risk

$$\text{Pecuniary Externality} = \mathbb{E} \left[\frac{u'(c)}{\mathbb{E}[u'(c)]} \cdot \left(\sum_{g=1}^G y_g \cdot \frac{dp_g}{d\mu} \right) \right]$$

Why does pecuniary externality reduce consumption risk?

- Under incomplete insurance, marginal utility higher in the bad state
- Insurance subsidy $\Rightarrow$ more production of risky crop
- $\Rightarrow$ lower relative price of risky crop $\Rightarrow$ transfers resources to bad state

Note that $\mathbb{E} \left[\sum_{g=1}^G y_g \cdot \frac{dp_g}{d\mu} \right] = 0$, so no effect on average income

- Pecuniary externality is just about transfers *across* states through *relative* prices

Pecuniary Externalities Under Incomplete Markets

When and why are pecuniary externalities non-zero?

When separation result holds, can rewrite pecuniary externality:

$$\mathbb{E} \left[\frac{u'(c)}{\mathbb{E}[u'(c)]} \cdot \left(\sum_{g=1}^G y_g \cdot \frac{dp_g}{d\mu} \right) \right] = (\mu + \tau) \cdot \sum_{g=1}^G \left(y_g(s^L) - y_g(s^H) \right) \cdot \frac{dp_g}{d\mu}$$

- Holds for arbitrary production functions and (homothetic) crop demand
- Under full insurance, $\mu + \tau = 0$, the pecuniary externality is zero
- When markets are competitive ($\mu = 0$) and complete ($\tau = 0$), both fiscal and pecuniary externality are zero

Special case of more general principle:

- First Welfare Theorem: Competitive and complete markets achieve the first-best
 $\implies$ pecuniary externalities cancel out
- Theory of the Second Best: Away from first-best, they no longer cancel out!

Optimal Subsidies in Model with Extensive Margin

What happens when we add an extensive margin to insurance choice?

At the optimal subsidy:

$$\underbrace{-\mu \cdot \frac{dx}{d\mu}}_{\text{Fiscal Externality}} = \underbrace{\mathbb{E} \left[\frac{u'(c)}{\mathbb{E}[u'(c)]} \cdot \left(\sum_{g=1}^G y_g \cdot \frac{dp_g}{d\mu} \right) \right]}_{\text{Pecuniary Externality}} + \underbrace{\mathbb{E} \left[\frac{u'(c)}{\mathbb{E}[u'(c)]} \cdot (\mathbb{E}[x] - x) \right]}_{\text{Fiscal Incidence Term}}$$

- Subsidy, funded with lump-sum taxes, redistributes from households that don't buy insurance to households that do buy
- Fiscal incidence term measures effect of this redistribution on welfare
- In quantitative model, participants have higher expected marginal utility
 $\implies$ Fiscal incidence term pushes optimal subsidy lower

The Optimal Subsidy Implements a Negative Markup

What markup, μ^* , solves the optimal-subsidy FOC?

For model with no extensive margin:

- At $\mu = 0$, fiscal externality is zero; at $\mu + \tau = 0$, pecuniary externality is zero
- So, $\mu^* < 0$, but not all the way to full insurance
- Optimal subsidy undoes any markup, plus a bit more

For model with extensive margin:

- Fiscal incidence term can push against pecuniary externality
- Optimal markup is ambiguous: can have $\mu^* > 0$

Result highlights importance of macroeconomic analysis for optimal policy

- Pecuniary externality is inherently a general equilibrium object
- Micro RCTs inform calibration, but cannot directly tell us optimal subsidy

Calibration

Calibration Strategy

Our calibration strategy proceeds in three steps:

1. Externally calibrate certain parameters ($\sigma, \phi, \mu^{\text{Baseline}}, \tau$)
2. Use data from Ghana Living Standards Survey, plus rainfall data, to help identify production parameters
3. Simulate (general equilibrium) model for parameter guess ($\gamma, \chi, \bar{\varepsilon}, \theta^*$), and search to match (partial equilibrium) experimental moments from Karlan et al. (2014)

In the interest of time, will skim over 1 and 2, to focus on 3

Overview of Steps 1 and 2

Step 1: Externally calibrate certain parameters

- Elasticity of substitution across crops: $\sigma = 2.4$ from Sotelo (2020)
- Elasticity of Output with Respect to Fertilizer: $\phi = 0.4$ from Donovan (2021)
- Baseline Markup: $\mu^{\text{Baseline}} = 0$ by assumption
- Servicing Costs: $\tau = 0.4 \times P(s^L)$ to match $\frac{\text{Market Price}}{\text{Actuarially Fair Price}}$ (Karlan et al., 2014)

Step 2: Match GLSS and rainfall data

- Get data on yields and land shares from Ghana Living Standards Survey (GLSS)
- Focus on nine largest crops by area (exclude cocoa):
Maize, sorghum, groundnut (peanut), beans, rice, cassava, plantain, yam, cocoyam (taro)
- Combine with rainfall data from Google Earth Engine (UCSB-CHIRPS data)
- Moments identify $P(s^L)$ and $\{z_g(s_i)\}_{g=1}^G$, but only identify ratio of curvature parameters (e.g. $\theta_{\text{Maize}}/\theta_{\text{Rice}}$)
- Need one more moment to identify the system (Step 3)

Step 3: Matching Experimental Moments

Most parameters pinned down, just need to guess four $(\gamma, \chi, \bar{\varepsilon}, \theta^*)$

We want to simulate the Karlan et al. experiment within our model

- Randomly offer insurance to households at different prices

This is a partial-equilibrium experiment: change insurance price without changing other equilibrium objects

To simulate this in the model:

- Simulate model in general equilibrium for parameter guess
- Compute household behavior under different $p_{x,i}$, holding crop prices $\{p_g\}_{g=1}^G$ and insurer profits Π' constant
 - Treated households get actuarially fair price, control households get market price
- Update guess and iterate until we match experimental moments

Step 3: Insurance Demand Moments

In principle, parameters jointly identified from all moments

- But, some sense of connection between specific moments and parameters

Insurance Demand Moments:

- Take-up Probabilities: 20% at market price, 40% at actuarially fair
 - These identify participation cost parameters, $(\chi, \bar{\varepsilon})$
- Arc Elasticity of Demand: -4.1 from market to actuarially fair price
 - Conditional on take-up, households will fully insure at fair price, while demand at market price is decreasing in γ
 - Higher $\gamma \iff$ Less elastic demand

We estimate $\gamma = 4.4$, $\chi = 0.16$, and $\bar{\varepsilon} = 8.7$

Demand moments are the key inputs for fiscal externality and fiscal incidence

Step 3: Effect of Insurance on Crop Riskiness

Finally, need one more moment to identify the curvature parameters $\{\theta_g\}_{g=1}^G$

Effect of Insurance:

- Karlan et al. use price offers as instrument for effect of insurance take-up
 - *Note: Instrument does not satisfy exclusion restriction, but still a moment we can match*
- One outcome is land share allocated to maize, which is risky crop in their setting
 - Increases by 9 percentage points, from baseline of 31%
- In Ghana as a whole, maize is not especially risky, so we select the three riskiest crops in our data: groundnut, rice, and cassava
 - Make up 28.3% of land at baseline; target a one-third increase in land share

Step 3: Effect of Insurance on Crop Riskiness (Continued)

We want to replicate IV estimate of insurance on crop choice

- Implement as Wald estimator:

$$\beta^{IV} = \frac{\mathbb{E} \left[\sum_{g \in \text{Risky Crops}} l_{g,i} \mid \text{Treated} \right] - \mathbb{E} \left[\sum_{g \in \text{Risky Crops}} l_{g,i} \mid \text{Control} \right]}{P(x_i > 0 \mid \text{Treated}) - P(x_i > 0 \mid \text{Control})}$$

- The curvature parameter governs elasticity of land share w.r.t. productivity shock
 - Insurance subsidy makes risky crops more productive in risk-adjusted sense

Weighted average θ is 0.62 (using land shares as weights)

Response of crop mix to subsidy is key input to pecuniary externality

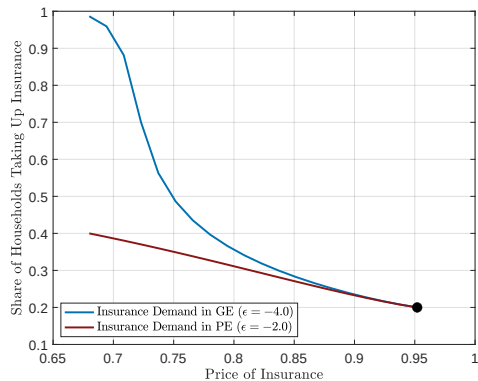
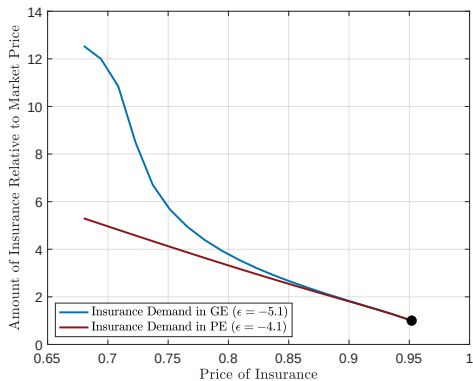
Calibration Table

Parameter	Description	Value	Calibration Target
σ	Elasticity of Substitution Across Crops	2.4	Sotelo (2020)
ϕ	Elasticity of Output w.r.t. Fertilizer	0.4	Donovan (2021)
μ^{Baseline}	Markup on Insurance	0	By Assumption
τ	Servicing Cost of Insurance	$0.4 \times P(s^L)$	Computed in Karlan et al. (2014)
$P(s^L)$	Probability of Low Rainfall	0.68	Rainfall Data
$\{z_g(s_i)\}_{g=1}^G$	State-Dependent Productivity by Crop	—	Yield Regressions and Land Shares
$\{\theta_g\}_{g=1}^G$	Curvature Parameter by Crop	—	Yields and Effect of Insurance
γ	Coefficient of Relative Risk Aversion	4.4	Elasticity of Demand for Insurance
$(\chi, \bar{\varepsilon})$	Participation Cost Parameters	(0.16, 8.7)	Insurance Take-Up Probabilities

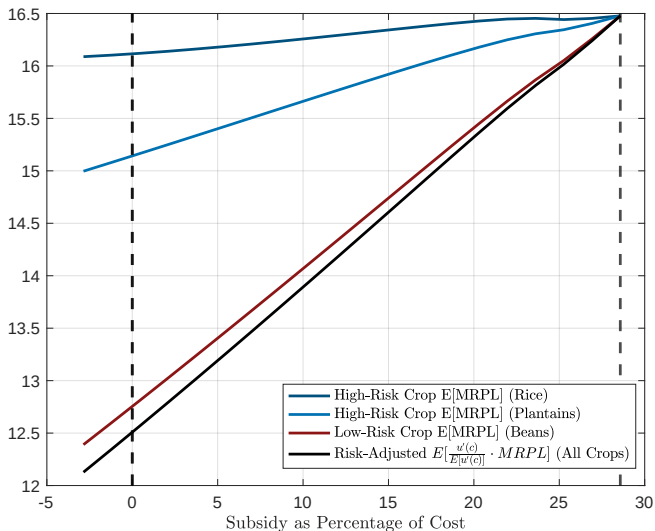
Table 1: Calibration Table for Model Parameters

Results and Counterfactuals

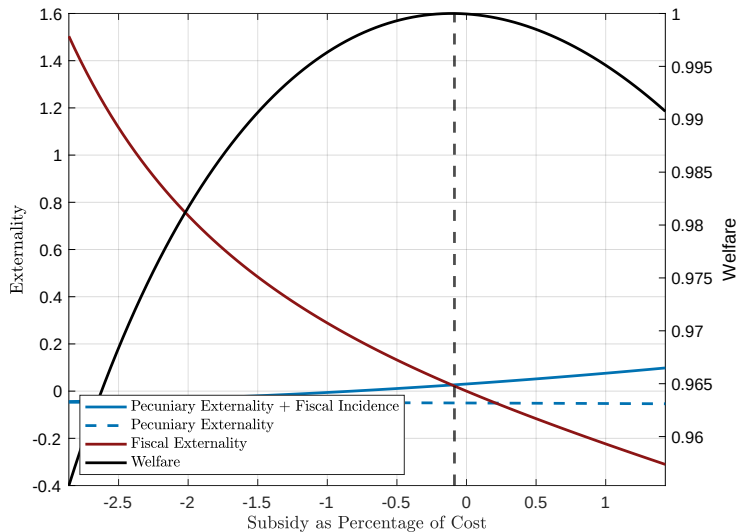
Insurance Demand (Partial Equilibrium vs. General Equilibrium)



Marginal Revenue Product of Land (Among Households that Buy Insurance)



The Optimal Subsidy is Close to Zero: -0.09% of Cost of Insurance



How Does Risk Aversion Affect Optimal Subsidy?

Insurance is valuable because households are risk averse

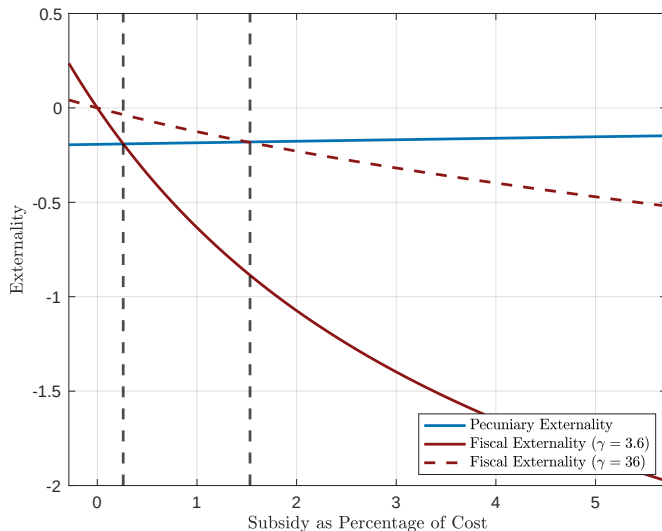
When risk aversion is higher, insurance is more valuable

How does the level of risk aversion affect the optimal subsidy?

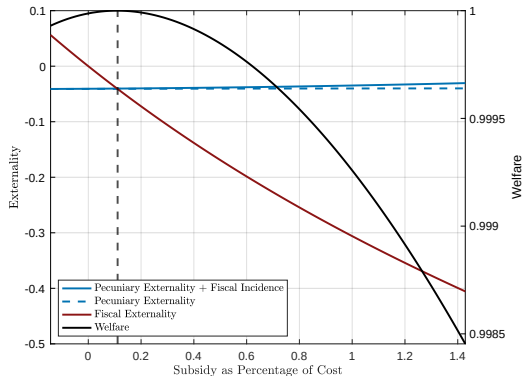
- Insurance is more valuable, but this is internalized by household
- What matters is how risk aversion affects the fiscal externality, pecuniary externality, and the fiscal incidence term

We will start by analyzing the model with no extensive margin

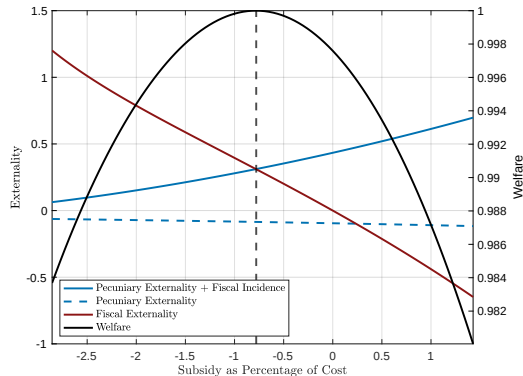
With No Extensive Margin, Optimal Subsidy Increases with Risk Aversion



With Extensive Margin, Comparative Static is Ambiguous



$\gamma = 3$, Optimal Subsidy = 0.11%



$\gamma = 8$, Optimal Subsidy = -0.78%

Why is the Optimal Subsidy Small?

Across calibrations, optimal subsidy is consistently small... Why?

Insurance demand is elastic, so fiscal externality grows quickly away from zero

- Fiscal externality will pull the optimal subsidy back towards zero

Pecuniary externality is small

- Perhaps because it is a very indirect benefit of insurance

Fiscal incidence term pushes against the pecuniary externality

- Overpowers the pecuniary externality in calibrated model with extensive margin

Markets are incomplete, but insurance subsidies do not undo the friction

Conclusion

Should we subsidize agricultural insurance?

- Built a general equilibrium model of agricultural households with risk, investment, crop choice, and insurance
- Showed optimal subsidy balances fiscal externality, pecuniary externality, and fiscal incidence term
- Calibrated model to match the data, including experimental estimates
- Find that the optimal subsidy is close to zero: -0.09%.

More broadly, highlight valuable back-and-forth between micro and macro

- Macroeconomic modeling is essential to study optimal policy
 - Especially when incomplete markets $\implies$ pecuniary externalities
- Microeconomic experiments are very valuable to calibrate the model
 - Ideally, combine both!

Appendix

Identifying Production Parameters

For each crop, need to identify three production function parameters:

- Curvature (returns-to-scale) parameter θ_g
- Productivity, $z_g(s_i)$, in good state and bad state

We identify these using data on land use and yields

First, suppose you know θ_g :

$$z_g(s_i) = y_{g,i}(s_i) / l_{g,i}^{\theta_g}$$

Need data on land shares for each crop, and (state-specific) output

Subtlety: We measure revenue, rather than output.

- Back out p_g from $\mathbb{E}[p_g \cdot y_g]$, then can back out $y_g(s_i)$
- Normalization: Demand has no crop-specific multipliers, so need to infer “units”

Identifying Production Parameters (θ_g)

How do we identify curvature parameter θ_g ?

Risk-neutral farmers would equalize marginal revenue product of land across crops

$$\begin{aligned}\frac{d}{dl_g} \mathbb{E}[p_g y_{g,i}] &= \theta_g \cdot p_g \cdot \mathbb{E}\left[z_g \cdot l_{g,i}^{\theta_g-1}\right] \\ &= \theta_g \cdot \underbrace{\frac{p_g \cdot \mathbb{E}[y_{g,i}]}{l_g}}_{\text{Average Yield for Crop } g}\end{aligned}$$

So, ratio of average yields gives ratio of θ_g parameters

With risk aversion, $\mathbb{E}[\cdot]$ replaced with risk-adjusted probabilities

- If demand for insurance is positive, this is pinned down by p_x

Ghana Living Standards Survey (GLSS) Data

Pin down production parameters with moments from GLSS

Data on revenue and land allocation by crop

- Revenue data available from rounds 4-7 (1998-1999, 2005-2006, 2012-2013, 2016-2017)
- Land data is only available for a subset of crops
- For round 7, we are able to impute area harvested; verify accuracy for the crops where we observe land directly

We focus on the top ten crops by area harvested, and then drop cocoa

- Maize, sorghum, groundnut (peanut), beans, rice, cassava, plantain, yam, cocoyam (taro)

Rainfall Data

Daily rainfall data from Google Earth Engine (UC Santa Barbara CHIRPS data):

- Data from 1981-2025, with 0.05 degree (≈ 5.5 km) pixel resolution
- For each pixel, compute “Bad Weather” dummy for a year

Following exact definition used in Karlan et al.’s Year 2 insurance product

- Growing season from June-September
- Day is “dry” if less than 1mm rain, and “wet” if more than 1mm
- Bad weather trigger if 12 or more consecutive dry days and/or 7 or more consecutive wet days

Take average to get $P(s^L) = 0.68$

Productivity Regressions

To back out average yields, we run (in round 7):

$$\log(\text{Yield}_{irg}) = \alpha_r + \delta_g + \varepsilon_{irg}$$

To back out how yields depend on weather, we regress on Bad Weather_{irt} (rounds 4-7)

- Bad Weather_{irt} variable is average across all pixels in that region

For crops where we can observe yields:

$$\log(\text{Yield}_{irgt}) = \beta_g \cdot \text{Bad Weather}_{irt} + \gamma_{rg} + \theta_{gt} + \varepsilon_{irgt}$$

For crops where we cannot observe yields:

$$\log(\text{Harvest Value}_{irgt}) = \beta_g \cdot \text{Bad Weather}_{irt} + \gamma_{rg} + \theta_{gt} + \varepsilon_{irgt}$$

where i is a household, r is a region, g is a crop, t is a year

Imputation of Land in GLSS

GLSS divides crops into two groups, based on typical harvest frequency

- For frequently harvested crops, does not ask household for total acreage
- But, we have total harvest value for all crops

In round 7, we have area of plot, and all crops harvested on it (may be multiple)

Imputation procedure:

- Trim harvest value by crop (1st and 99th percentile)
- For each region-crop, compute yield among single-cropped plots
- Divide harvest value by relevant imputed yield to get acreage

Confirm accuracy for the crops where we observe actual acreage [Back](#)

Data Cleaning for GLSS Regressions

For productivity regressions:

- Trim harvest value by crop-year at 1st and 99th percentile
- Compute yields (actual or imputed acreage)
- Then trim yields by crop-year at 1st and 99th percentile

For crop fixed effect regressions use round 7 yields

For rainfall regressions use rounds 4-7

- Use yields for crops where we can compute them
- Harvest value is the outcome for other crops

GLSS Estimates

Crop	Acreage	Imputed Acreage	δ_g	β_g
Beans	2,815.74	3,093.89	0.00	0.23
Cassava		4,907.58	0.51	-0.66
Cocoyam		2,557.52	0.06	-0.65
Groundnut	4,031.86	4,916.80	0.56	-0.51
Maize	11,053.42	11,691.24	0.37	-0.21
Plantain		2,960.37	0.55	-0.59
Rice	2,061.25	3,037.53	0.69	-0.37
Sorghum	5,542.16	4,184.56	-0.04	-0.21
Yam		2,889.68	1.66	-0.53

Table 2: Estimated Moments from Ghana Living Standards Survey

Calibration of Productivity Parameters (Details)

Calibrate model to match:

- $\frac{p_g \cdot \mathbb{E}[y_{g,i}]}{l_g} = \exp(\delta_g)$
- $\frac{p_g \cdot y_{g,i}(s_i^L)}{l_g} = \exp(\beta_g) \cdot \frac{p_g \cdot y_{g,i}(s_i^H)}{l_g}$
- $\mathbb{E}[l_g]$ from land shares in round 7

Separation means we can solve for production decisions without solving household problem

- Solve for $p_x = P(s^L) + \tau$, since $\mu^{\text{Baseline}} = 0$
- Solve land FOCs to get θ_g from δ_g and β_g
- Use σ to get p_g from δ_g
- Solve for state-specific productivities
- Everything is closed-form :-)